LEON NSAMBA GRADIENT DESCENT

$y = m_1x_1 + m_2x_2 + b$

$\hat{y} = mx + b$, $m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $m_1 = -1$, $m_2 = 2$, $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$, $x_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $x_2 = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$,

$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

x	y	$\hat{y}$	$\hat{y} - y$
$\begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$	$\begin{bmatrix} 5 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 6 \\ 17 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Getting $\hat{y}$ (yhat)

$\hat{y} = mx + b$

$\hat{y} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} (-1 \times 1) + (2 \times 3) \\ (-1 \times 4) + (2 \times 10) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$

$\hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

$\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Calculating the ~~loss~~ MSE

$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2$, $n = 2$

$mse = \frac{1}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix}^2 \rightarrow \frac{1}{2} \left(\begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 11 \end{bmatrix} \right)$

$MSE = \frac{1}{2} \begin{bmatrix} (1 \times 1) + (11 \times 11) \end{bmatrix}$

$MSE = \frac{1}{2} \times 122$

$MSE = 61$

Calculating the loss function(s)

Let MSE be J

$$\frac{dJ}{dm_1} = ?, \quad \frac{dJ}{dm_2} = ?, \quad \frac{dJ}{db} = ?$$

From

$$J = \frac{1}{n}\sum^{n}_{i=1} (\hat{y} - y)^2$$

But

$$\hat{y} = m_1x_1 + m_2x_2 + b$$

So

$$J = \frac{1}{n}\sum^{n}_{i=1} ((m_1x_1 + m_2x_2 + b) - y)^2$$

let

$m_1x_1 + m_2x + b - y$ be e

$$J = \frac{1}{n}\sum^{n}_{i=1} e^2$$

→ Applying chain rule

$$\frac{dJ}{dm_1} = \frac{dJ}{de} \cdot \frac{de}{dm_1}$$

$$\frac{dJ}{de} = \frac{2}{n}\sum^{n}_{i=1} e$$

$$e = m_1x_1 + m_2x_2 + b - y$$

$$\frac{de}{dm_1} = f(x_1 + 0 + 0)$$

m_2, x_2, b, y behave as constants and go upon differentiation (become 0)

$$\frac{de}{dm_1} = x_1 + 0 + 0 - 0$$

$$\frac{de}{dm_1} = x_1$$

$$\frac{dJ}{dm_1} = \frac{2}{n}\sum^{n}_{i=1} e \cdot x_1$$

Since $e = (m_1x_1 + m_2x_2 + b - y)$, with $\hat{y}$ and y

$$\frac{dJ}{dm_1} = \frac{2}{n}\sum^{n}_{i=1} (\hat{y} - y)x_1 \quad \text{—— (1)}$$

$\frac{dJ}{dm_2} = ?$

$$\boxed{\frac{dJ}{dm_2} = \frac{dJ}{de} \cdot \frac{de}{dm_2}} \rightarrow \text{Applying chain rule}$$

$$J = \frac{1}{n}\sum_{i=1}^{n} (\hat{y} - y)^2$$

But $\hat{y} = m_1 x_1 + m_2 x_2 \overset{+b}{\cancel{y}}$

let $e = \hat{y} - y$

$$e = m_1 x_1 + m_2 x_2 \overset{+b}{\hat{}} - y$$

$$J = \frac{1}{n}\sum_{i=1}^{n} e^2$$

$$\boxed{\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e}$$

$$e = m_1 x_1 + m_2 x_2 \overset{+b}{\hat{}} - y$$

$$\frac{de}{dm_2} = 0 + x_2 + 0 - 0$$

m_1, x_1, b, y are constants and become 0 upon differentiation

$$\boxed{\frac{de}{dm_2} = x_2}$$

$$\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} e \cdot x_2$$

$$\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} (\underbrace{m_1 x_1 + m_2 x_2 \overset{+b}{\hat{}}}_{\hat{y}} - y)\, x_2$$

$$\boxed{\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)\, x_2} \quad \text{——— (11)}$$

$\frac{dJ}{db} = ?$

$J = \frac{1}{n}\sum_{i=1}^{n} e^2$, where

$e = m_1 x_1 + m_2 x_2 + b - y$

$e = \hat{y} - y$

$\frac{dJ}{db} = \frac{dJ}{de} \cdot \frac{de}{db}$ → Applying chain rule

$\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e$

$\frac{de}{db} = 0 + 0 + 1 - 0$

$\frac{de}{db} = 1$

$\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} e \cdot 1$ where $e = \{m_1 x_1 + m_2 x_2 + b\} - y$ ($\{\cdots\} = \hat{y}$)

$\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)$ — (iii)

Therefore

$\frac{dJ}{dm_1} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y) x_1$, $\frac{dJ}{dm_2} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y) x_2$, $\frac{dJ}{db} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)$

$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $m_1 = -1, m_2 = 2$

$\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$,

$x_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $x_2 = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$

$n = 2$

$$m_1 new = m_1 old - \alpha \frac{dJ}{dm_1}$$

$$m_1 new = m_1 old - \alpha \left[\frac{+2}{n} \sum_{i=1}^{n} (\hat{y} - y) x_1 \right]$$

$$m_1 new = -1 - \alpha \left[\frac{+2}{2} \sum_{i=1}^{n} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 4 \end{bmatrix} \right]$$

$$m_1 new = -1 - \alpha \left[+1 \ (1 \times 1) + (11 \times 4) \right]$$

$$m_1 new = -1 - \alpha \left[+45 \right]$$

$$\boxed{m_1 new = -1 - 45\alpha} \quad \text{—— (i)}$$

$$m_2 new = m_2 old - \alpha \left[\frac{2}{n} \sum_{i=1}^{n} (\hat{y} - y) x_2 \right]$$

$$m_2 new = 2 - \alpha \left[\frac{2}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \begin{bmatrix} 3 \\ 10 \end{bmatrix} \right]$$

$$m_2 new = 2 - \alpha \left[(1 \times 3) + (11 \times 10) \right]$$

$$m_2 new = 2 - \alpha \left[113 \right]$$

$$\boxed{m_2 new = 2 - 113\alpha} \quad \text{—— (ii)}$$

$b = 1$

$$b_{new} = b_{old} - \alpha \frac{dJ}{db}$$

$$b_{new} = b_{old} - \alpha \left[\frac{3}{n} \sum_{i=1}^{n} (\hat{y} - y) \right]$$

$$b_{new} = b_{old} - \alpha \left[\frac{3}{2} \begin{bmatrix} 1 \\ 11 \end{bmatrix} \cdot 1 \right]$$

$$b_{new} = 1 - \alpha \left[(1 \times 1) + (11 \times 1) \right]$$

$$\boxed{b_{new} = 1 - 12\alpha} \quad (1)$$

$$\boxed{m_{1new} = -1.45\alpha, \; m_{2new} = 2 - 113\alpha, \; b_{new} = 1 - 12\alpha}$$

Let $\alpha = 0.01$

$m_{1new} = -1.45(0.01)$

$m_{1new} = -1.45$

$m_{1new} = -0.55 - 1.45$

$m_{2new} = 2 - 113\alpha$

$m_{2new} = 2 - 113(0.01)$

$m_{2new} = 0.87$

$b_{new} = 1 - 12\alpha$

$b_{new} = 1 - 12(0.01)$

$b_{new} = 0.88$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2$$

$\hat{y} = mx + b$

$$\hat{y} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} (-1.45 \times 1) + (-1.45 \times 3) \\ (0.87 \times 4) + (0.87 \times 10) \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} -5.8 \\ 12.18 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

MSE when $m_1 = -1.45$, $m_2 new = 0.87$, $b new = 0.88$

$\hat{y} = mx + b$ $\quad m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$

$$\hat{y} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

$$\hat{y} - y = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2$$

$$MSE = \frac{1}{2} \left[(-2.96)^2 + (-2.22)^2 \right]$$

$$MSE = 6.85$$

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Daniel GAKUMBA NDWALI

Iteration 2.

$m = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix}$ $b = \begin{pmatrix} 0.88 \\ 0.88 \end{pmatrix}$ $X = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix}$ $y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$

Let us continue considering $MSE = J$

$y = m_1 x_1 + m_2 x_2 + b$, & $e = \hat{y} - y$, $J = (1/n)\Sigma e^2$

$dJ/dm = (2/n) X^T e$ $dJ/db = (2/n)\Sigma e_i$, $\alpha = 0.01$

$\hat{y} = X.m + b$, $m_{new} = m_{old} - \alpha\, dJ/dm$; $b_{new} = b_{old} - \alpha\, dJ/db$

& Let us calculate $X.m$

$$X.m = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} = \begin{pmatrix} (1)(-1.45) + (3)(0.87) \\ (4)(-1.45) + (10)(0.87) \end{pmatrix} = \begin{pmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{pmatrix}$$

$$X.m = \begin{pmatrix} 1.16 \\ 2.90 \end{pmatrix}$$

now $\hat{y} = X.m + b$

$$\hat{y} = \begin{pmatrix} 1.16 \\ 2.90 \end{pmatrix} + \begin{pmatrix} 0.88 \\ 0.88 \end{pmatrix} = \begin{pmatrix} 1.16 + 0.88 \\ 2.90 + 0.88 \end{pmatrix}$$

$$\hat{y} = \begin{pmatrix} 2.04 \\ 3.78 \end{pmatrix}$$

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Error : $E = \hat{y} - y$

$$E = \begin{pmatrix} 2.04 \\ 3.78 \end{pmatrix} - \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 2.04 - 5 \\ 3.78 - 6 \end{pmatrix}$$

$$e = \begin{pmatrix} -2.96 \\ -2.22 \end{pmatrix}$$

now Let us find MSE = J.

$$J = \frac{1}{n}(e_1^2 + e_2^2)$$

$$J = \frac{1}{2}\left((-2.96)^2 + (-2.22)^2\right) = \frac{1}{2}(8.7616 + 4.9284) = \frac{13.69}{2}$$

$$J = 6.845$$

now Let us find the gradient for $m \Rightarrow (2/n) X^T e = dJ/dm$

where $X^T = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}$

$$dJ/dm = (2/n) X^T e = \left(\frac{2}{2}\right)\begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}\begin{pmatrix} -2.96 \\ -2.22 \end{pmatrix} = \begin{pmatrix} (1)(-2.96) + (4)(-2.22) \\ (3)(-2.96) + (10)(-2.22) \end{pmatrix} = \begin{pmatrix} -2.96 - 8.88 \\ -8.88 - 22.20 \end{pmatrix}$$

$$2/n\, X^T e = \begin{pmatrix} -11.84 \\ -31.08 \end{pmatrix} = dJ/dm$$

now Let us find the gradient for $b \Rightarrow (2/n)\Sigma e = dJ/db$.

$$dJ/db = (2/n)\Sigma e = \left(\frac{2}{2}\right)\left((-2.96) + (-2.22)\right) = -2.96 - 2.22$$

$$2/n \Sigma e = -5.18 = dJ/db.$$

now Let us update and have m_{new} and b_{new}

$$m_{new} = m_{old} - \alpha \, dS/dm$$

$$m_{1\,new} = -1.45 - [0.01(-11.84)] = -1.45 + 0.1184$$

$$m_{1\,new} = -1.3316$$

$$m_{2\,new} = 0.87 - [0.01(-31.08)] = 0.87 + 0.3108$$

$$m_{2\,new} = 1.1808$$

$$b_{new} = b_{old} - \alpha \, dS/db$$

$$b_{new} = 0.88 - 0.01(-5.18) = 0.88 + 0.0518 =$$

$$b_{new} = 0.9318$$

Michael Odhiambo Iteration 3

Given

$$m = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}, \quad b = \begin{pmatrix} 0.9318 \\ 0.9311 \end{pmatrix} \quad x = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \quad y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

Considering MSE =]

$$y = m_1 x_1 + m_2 x_2 + b, \quad e = \hat{y} - y, \quad J = \left(\frac{1}{n}\right) \sum e^2$$

$$\frac{dJ}{dm} = \left(\frac{2}{n}\right) X^T e \qquad \frac{dJ}{db} = \frac{2}{n} \sum e, \quad \alpha = 0.01$$

$$\hat{y} = \alpha \cdot m + b, \quad m_{new} = m_{old} - \alpha \frac{dJ}{dm}; \quad b_{new} = b_{old} - \alpha \frac{dJ}{db}$$

Calculate $x \cdot m$

$$x \cdot m = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.3316 \\ 1.1808 \end{pmatrix} = \begin{pmatrix} 1(-1.3316) + 3(1.1808) \\ 4(-1.3316) + 10(1.1808) \end{pmatrix} = \begin{pmatrix} 2.2108 \\ 6.4816 \end{pmatrix}$$

Prediction $\hat{y} = x \cdot m + b$

$$\hat{y} = \begin{pmatrix} 2.2108 \\ 6.4816 \end{pmatrix} + 0.9318 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2.2108 \\ 6.4816 \end{pmatrix} + \begin{pmatrix} 0.9318 \\ 0.9318 \end{pmatrix} = \begin{pmatrix} 3.1426 \\ 7.4134 \end{pmatrix}$$

error $E = \hat{y} - y$

$$E = \begin{pmatrix} 3.1426 \\ 7.4134 \end{pmatrix} - \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -1.8574 \\ 1.4134 \end{pmatrix}$$

M.S.E $J \to \frac{1}{n} E^T E$

$$= \frac{1}{2} \begin{pmatrix} -1.8574 & 1.4134 \end{pmatrix} \begin{pmatrix} -1.8574 \\ 1.4134 \end{pmatrix}$$

$$= \frac{1}{2} (-1.8574)^2 + (1.4134)^2$$

$$= \frac{1}{2} (5.4476)$$

$$= 2.7288$$

Gradient for m $\frac{dJ}{dm} = \frac{2}{n} X^T E \qquad x = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}$

Gradient for b $\frac{\partial J}{\partial b} = 2/n\ I^T E$

$= \frac{2}{2}\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}-1.8874\\ 1.4134\end{pmatrix}$

$= \frac{2}{2}\left(1(-1.8874) + 1(1.4134)\right)$

$= \frac{2}{2}\left(-1.8874 + 1.4134\right)$

$= -0.4440$

Continuation gradient for m

$= \frac{2}{2}\begin{pmatrix}1 & 4\\ 3 & 10\end{pmatrix}\begin{pmatrix}-1.8574\\ 1.4134\end{pmatrix}$

$= \begin{pmatrix}1(-1.8574) + 4(1.4134)\\ 3(-1.8574) \quad 10(1.4134)\end{pmatrix}$

$= \begin{pmatrix}-1.8574 + 5.6536\\ -5.5722 + 14.1340\end{pmatrix}$

$= \begin{pmatrix}3.7962\\ 8.5618\end{pmatrix}$

Update m $\quad m_{new} = m_{old} - \alpha \frac{\partial J}{\partial m}$

$= \begin{pmatrix}-1.3316\\ 1.1808\end{pmatrix} - 0.01\begin{pmatrix}3.7962\\ 8.5618\end{pmatrix} = \begin{pmatrix}1.3316\\ 1.1808\end{pmatrix} - \begin{pmatrix}0.0380\\ 0.0856\end{pmatrix} = \begin{pmatrix}-1.3696\\ 1.0952\end{pmatrix}$

Update b $\quad b_{new} = b_{old} - \alpha \frac{\partial J}{\partial m}$

$= 0.9318 - 0.01(-0.4440) \rightarrow 0.9318 + 0.0044 = 0.9362$

MSE after the update

$\hat{y} = x.m_{new} + b_{new}$

$= \begin{pmatrix}1 & 3\\ 4 & 10\end{pmatrix}\begin{pmatrix}-1.3696\\ 1.0952\end{pmatrix} + 0.9362\begin{pmatrix}1\\ 1\end{pmatrix}$

$= \begin{pmatrix}1(-1.3696 + 3(1.0952)\\ 4(-1.3696 + 3(1.0952)\end{pmatrix} + \begin{pmatrix}0.9362\\ 0.9362\end{pmatrix}$

$= \begin{bmatrix}1.9160\\ 5.4736\end{bmatrix} + \begin{pmatrix}0.9362\\ 0.9362\end{pmatrix}$

$= \begin{pmatrix}2.8522\\ 6.4098\end{pmatrix}$

$E = \hat{y} - y = \begin{bmatrix}2.8522\\ 6.4098\end{bmatrix} - \begin{pmatrix}5\\ 6\end{pmatrix} = \begin{pmatrix}2.1478\\ 0.4098\end{pmatrix}$

$MSE = \frac{1}{2}\left((-2.1478)^2 + (0.4098)^2\right.$

$= \frac{1}{2}(4.6130 + 0.1679)$

$= 2.3905$

$\therefore m_1 = -1.3696$

$m_2 = 1.0952$

$b = 0.9362$

MSE update $= 2.3905$